What Each Agent Actually Does

12 stages, one pipeline, plain language. HVB-100 used as the running example throughout.

01

Demand Agent

Guesses how much of each product you'll need to make soon.

> Example: "Last 3 months, HVB-100 sold ~40 units/month, trending up" → outputs "forecast 45 units for HVB-100 next month."

Single-shot No tools — just reads raw sales history, outputs a number.

02

S&OP Agent

Checks: can we actually make what Demand wants, roughly?

> Example: Demand says 45 units of HVB-100. S&OP checks total plant capacity isn't already overcommitted that month.

Single-shot

03

MPS Agent (Master Production Schedule)

Turns the monthly plan into exact weekly build quantities.

> Example: "45 units this month" → "15 units week 1, 15 week 2, 15 week 3."

Single-shot

04

BOM Agent (Bill of Materials)

Looks up: what parts actually go into this product?

> Example: HVB-100 needs 1 valve body casting + 2 seals + 4 bolts. It just fetches this — doesn't invent it.

Single-shot → calls `routing-service`

05

MRP Agent (Material Requirements Planning)

Checks if you have enough raw material — and decides what to do if not.

> Example: Need 45 seals. Check stock → only 20 left → *decides* to also check the backup supplier before flagging a shortage. That "check, then decide next step" is why this one thinks in a loop, not one shot.

ReAct loop → calls `inventory-service`

06

Procurement Agent

Drafts a purchase order for whatever's short.

> Example: "Buy 500 more seals from Vendor X, needed by next Friday." Real money — this is one of the stages you'd actually want to eyeball before it goes further.

Single-shot

07

APS Scheduling Agent

Doesn't schedule anything itself — hands it to your real scheduler and explains the result.

> Example: calls the actual working scheduling-service you already built and tested. If it can't fit something, it explains why in plain English instead of a cryptic log line.

Single-shot → calls `scheduling-service` (the real engine, not a guess)

08

Shop Floor Agent

Reports what's actually happening on the machines right now.

> Example: "Op 3 of HVB-100's routing started 10 min late on CNC Mill 1." Just reporting reality, not deciding anything new.

Single-shot

09

Quality Agent

Records inspection results, flags anything that failed.

> Example: "Batch of 45 HVB-100 units: 43 passed, 2 flagged for rework."

Single-shot

10

Warehouse Agent

Decides where finished goods get stored.

> Example: "43 units of HVB-100 → shelf B12."

Single-shot

11

Dispatch Agent

Plans how finished goods get shipped out.

> Example: "43 units ready → schedule pickup with carrier for Thursday."

Single-shot

12

Analytics Agent

Looks back and summarizes how the whole run went.

> Example: "On-time: 95%. One shortage caused a 2-day delay on HVB-100." Nothing to approve — it's just the recap.

Single-shot

Every stage above writes its output to the database, always — even if nothing pauses for approval. That log is how you'd ever answer "where did it go wrong."